

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

How do ASP.NET Components Work?



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Represent the solution outline to the problem presented by the customer through making case of use diagrams, facing the problem to the customer following rules and protocols of the organization.

Ask your partner:

- What is ASP.NET?
- Why is ASP.NET important for web developers?

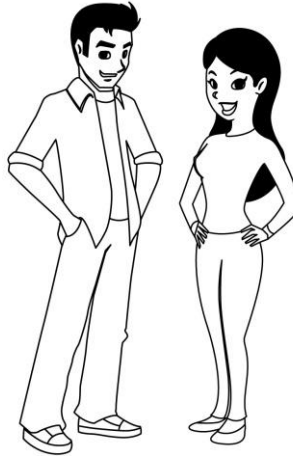
Can Do Statements:

- I can talk about the ASP.NET component model.
- I can list the components of .Net Framework 3.5.
- I can present some characteristics of components of .Net Framework 3.5.



Let's Explore!

James is doing an oral presentation about .NET Framework in his computer programming class.



Instructions: Read the text below.

- James:** The ASP.NET is part of the .NET framework. This system is based on object-oriented programming and it is made of pages.
- Katy:** Do you mean ASP.NET uses a component object model?
- James:** That's right! There are many pages in an ASP.NET web application, and these pages are made of an object-oriented hierarchy of classes in which each variable inherits features from another one.
- Katy:** How does this component model work?
- James:** When a user wants to have access to a page, he sends a request to a server or IIS; the server processes the information, turns it into HTML code and then, the server assigns the rest of the processing stage to the ASP. NET runtime system.
- Katy:** What does the runtime system do?
- James:** The runtime system gets the .aspx page and changes it into a .NET framework class. This is the result of inheritance process.
- Katy:** How many versions of .Net framework exist?
- James:** There are many. If I remember correctly, there are eight, but the one I know better is .Net Framework 3.5.
- Katy:** What is it made of? I mean, what components does it have?
- James:** Well, it has components such as common language runtime or CLR, metadata and assemblies, Windows presentation foundation, Windows CardSpace, among others.
- Katy:** What features do these components have?
- James:** Well, CLR offers services such control management and security checking. It analyses the code to execute programs and it normally uses just-in-time compilation. Windows presentation foundation is a tool to develop better visual features in an interface; and it is made of Windows applications that allow using media, animations, and files.
- Katy:** What about Windows CardSpace?
- James:** It is a technology to do identity control when we share personal data or download files.



**Let's Practice!**

She lists some components of .Net Framework 3.5.

Instructions: Read the statement. Select all the correct answers.

1. Net Framework 3.5. is made of:
 - a. CLR
 - b. Metadata
 - c. Services
 - d. Windows presentation foundation
 - e. Webpages

**Let's Practice!**

She describes some characteristics of components of .Net Framework 3.5.

Instructions: Read the clues. Complete the crossword.

1. It offers services such control management and security checking, it analyses the code to execute programs and it usually uses just-in-time compilation.
 - a. COM
 - b. CLR
 - c. ISS
 - d. d. C++
2. It is a tool to develop better visual features in an interface; it is made of Windows applications that allow using media, animations, and files.
 - a. Windows office
 - b. Windows outlook
 - c. Widows Vista
 - d. Windows presentation foundation
3. It is a technology to do identity control when we share personal data or download files. It does not require a password from the user.
 - a. HTML
 - b. Object-oriented programming
 - c. Windows CardSpace
 - d. Metadata



Extension Activity

Let's present a component of Net. Framework 3.5

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the components of Net. Framework 3.5
- Choose the one you think is the most important.
- Prepare an oral presentation for your classmates talking about the component you chose.
- Present it to your classmates.

Self-Assessment

Yes

No

- I look for information about the components of Net. Framework 3.5
- I prepare an oral presentation about a component of Net. Framework 3.5
- I present it to my classmates.

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Target Words

- **runtime system** (noun): a mechanism to process computer language to execute a program
e.g. The server processes the information, turns it into HTML code and, then, the server assigns the rest of the processing stage to the ASP. NET runtime system.
- **metadata** (noun): binary information related to a program
e.g. It has components such as common language runtime or CLR, metadata and assemblies, Windows presentation foundation, Windows cardspace, among others.
- **just-in-time compilation** (noun): a process to transform an intermediate representation of a code to a computer language at the same time of a program execution
e.g. It analyses the code to execute programs and it usually uses just-in-time compilation.
- **object-oriented programming** (noun): a programming language model based on objects and data
e.g. The ASP.NET makes part of the .NET framework; this system is based on object-oriented programming.
- **object-oriented hierarchy of classes** (noun): a classification of different objects depending on inheritance properties
e.g. Many pages in an ASP.NET web application are made of an object-oriented hierarchy of classes in which each variable inherits features from another one.
- **iis** (acronym): internet information service; a group of web servers
e.g. When a user wants to have access to a page, he sends a request to a server or IIS.



Learning Points

Be Made of to Talk about Components of ASP.NET

You use the structure **be + made + of** to describe the nature or the components of an object.

You can use the structure **be + made + of** to talk about the components of ASP.NET. Remember that you have to conjugate the verb Be according to the subject.

- An ASP.NET web application **is made of** multiple pages.
- ASP.NET pages **are made of** hierarchal classes.

You can use this structure in affirmative, negative and interrogative statements:

- An ASP.NET web application **is made of** multiple pages.
- An ASP.NET web application **isn't made of** multiple pages.
- **Is** an ASP.NET web application **made of** multiple pages?

Examples

- *This system is based on object-oriented programming, and it is made of pages.*
- *There are many pages in an ASP.NET web application, and these pages are made of an object-oriented hierarchy of classes.*

Overview on Windows CardSpace

Windows CardSpace is a technology based on an identity metasystem to recognize the user identity on the internet without requesting a password when signing in to a website.

Windows CardSpace allows using identification cards by means of XLM based protocols to have access to websites that accept this technology. When a website requests user information, Windows CardSpace sends an encrypted identification card, the website analyses the information and once the data is confirmed, the user can login without using a password.

To avoid security issues and protect user privacy, Windows CardSpace only sends personal information to authorized sites and keeps track of the websites where it sends identification cards.

